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ORIGINAL ARTICLE

Rapid parametric design methods for shoe-last customization

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Abstract With the development of computer-aided design (CAD) technology and increasing demands of customized footwear, shoe-lasts are requested to be designed rapidly so as to speed-up the process of footwear manufacturing. Thus, this study presents a CAD system for shoe-last rapid customized design based on the piecewise reconstruction to realize the interactive deformation and separate/global shoe-last form reuse. First, piecewise remodeling method is proposed based on the multi-layer parametric definition and contour curves are extracted from the mesh. Then, five types of proper constraints to support surface manipulation are proposed, and the draft-driven deformation by the contour curve bending can realize the interactive local surface design in free angle of view. Finally, shoe-last styles can be saved and reused globally or separately to share design results between different shoe-lasts. Experimental examples show that customized shoe-lasts can be easily and rapidly generated by adopting the parametric design methods.

Keywords Shoe-last · Parametric design · Interactive deformation · Form reuse

1 Introduction

Shoe-last, a 3D mold used for stretching the shoe leather and applying the sole, is an indispensable tool in the modern footwear industry. Thus shoe-last design is a critical link in footwear industry, which determines the type and style of the corresponding shoe. With the improvement of modernization and the increasing demands of customized shoes, there is a great need to speed-up shoe last design and fulfill people's request to wear customized shoes [1, 2].

Conventionally, a shoe-last is designed by drawing 2D outlines according to foot feature dimensions, and it is carved manually into mold by experienced shoe-last makers. It needs a long period to generate a shoe-last by repeatedly adjusting. However, much time is saved by applying CAD and CAM technology recently in shoe-last design. With efficient foot data scanning technique [3, 4], key dimensions (e.g., foot-length or girth) can be accurately extracted from foot form [5, 6], which can avoid errors caused by eyes and hands in traditional manual measurement. These dimensions in the shoe-last can be used as parameters which enable the further design of customized shoe-last. The existing studies show that there are many approaches aiming at shoe-last customization by means of parameter adjusting. In some studies, a general idea of designing customized shoe-last is based on foot form analysis [7, 8] and fitting evaluation [9–11]. Researches on analyzing foot form show that foot characteristics differ by people [12] and there are relations between foot dimensions [13]. To certain shoe-last style, a base shoe-last is selected to match the custom's foot form with global grading and local deformation until it meets the requirements-of-fit evaluation [14–17]. In addition, if there is a standard shoe-last, it can be automatically adapted to a customized foot form with a segment-based adaptive modification method [18].

Foot form parameterization is the base of shoe-last customization [16, 17], and it is introduced into shoe-last surface deformation in some studies. A parameterized shoe-

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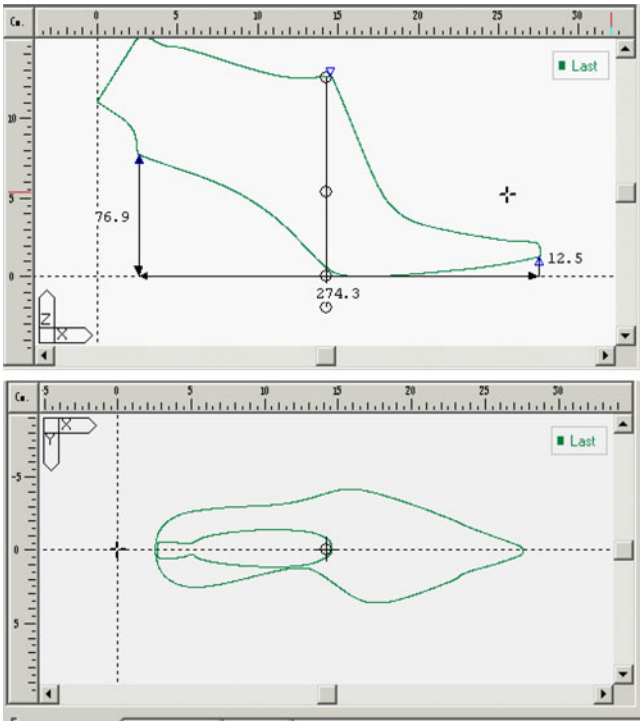


Fig. 1 EasyLast: shape modification by profile adjusting

last freeform deformation driven by nature product definition parameters is proposed in [19]; Kim et al. [20] introduced foot pressure unbalance analysis in addition to parameter-driven deformation and developed a conformal mapping approach for shoe last design. Luximon and Luximon [21] proposed a shoe-last design approach by parameter-driven toe or heel dimension adjusting based on a parametric shoe-last model. These methods still remain on simple dimension-driven surface deformation, which has limitation in interactive surface editing and cannot achieve intuitive surface manipulation. One of the most popular methods for complex surface design is to move control knots on non-uniform rational B-spline surface [22–24], but they do not take the features of shoe-last structure into consideration. Some commercial systems, such as Shoemaster and EasyLast, also provide the similar methods. However, these systems can only adjust shoe-last profile or outline in 2D method and lack of actually 3D interactive deformation (Figs. 1 and 2). Thus, we propose a innovate approach to realize the 3D interactive deformation by creating a contour curve frame from the features and modifying the surface in accordance with the curve bending. In addition, the existing studies seldom mention

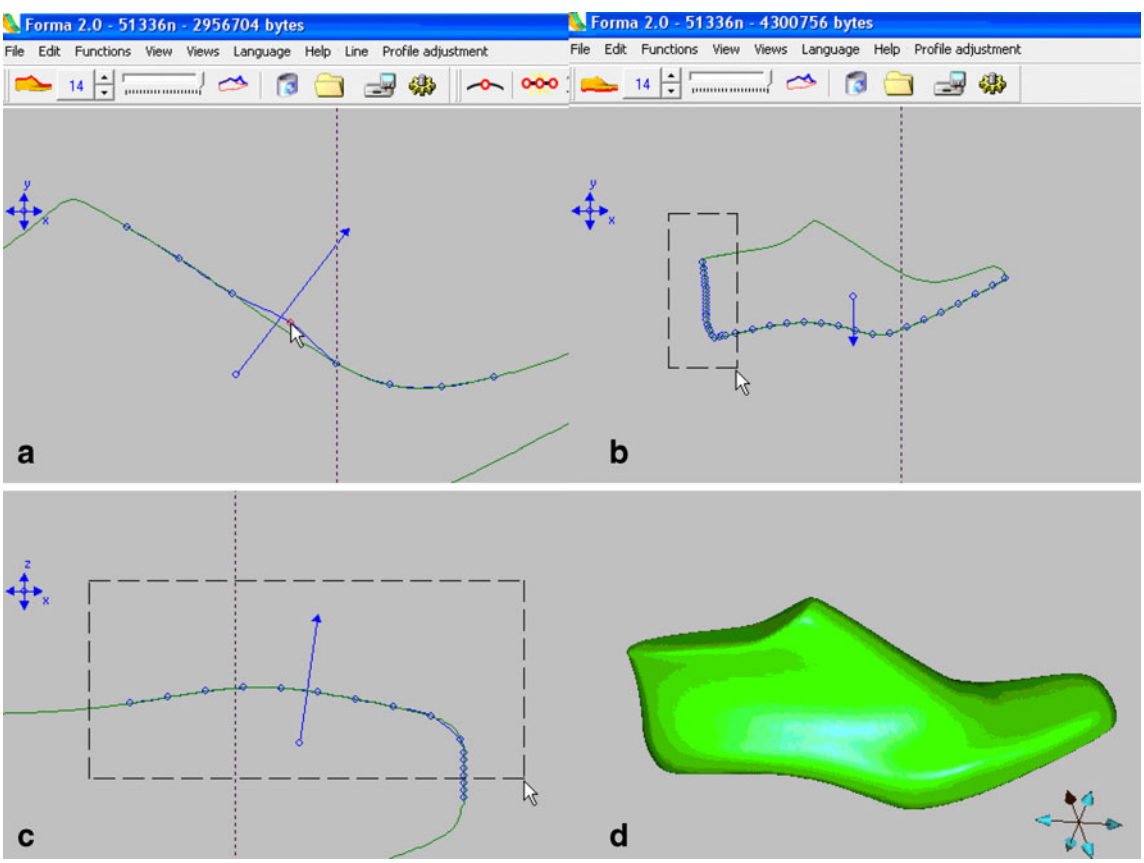


Fig. 2 Shoemaster: shape modification by adjusting centerline (**a**) and heel curve (**b**); shape modification by adjusting section girth parameters (**c**)

Table 1 MLPD of shoe-last

Layer	Element	Description
Feature point layer	$\{F_{\mathrm{pi}}\}$	Feature points for mesh construction, and contour curve extraction, etc.
Parametric curve layer	$\{S_{\mathrm{pli}}\}$	Spline curves for shoe-last deformation, which are established by piecewise surface boundaries.
Parametric surface layer	$\{M_{\mathrm{si}}\}$, M_w	$\{M_{\mathrm{si}}\}$ is the set of piecewise surfaces, while M_w is the whole shoe-last mesh.
Dimension layer	$\{S_i\}$	Dimensions of shoe-last length, height, girth, etc.
Constraint layer	$\{C_i\}$	Constraints keeping the robustness of the geometric and topologic structure of shoe-last (reference at Section 3.3).

shoe-last reuse methods which can speed-up design process and generate high quality model, and it needs further more research in the future [25].

Parametric model construction, constraint relations extraction, as well as solving method of constraints are the crucial steps in parametric design. We propose a multi-layer parametric definition (MLPD) which consists of feature point layer, parametric curve layer, parametric surface layer, etc., and obtain parametric piecewise reconstruction of shoe-last based on it. Then, contour curves are extracted from piecewise surfaces and parametric constraints are introduced into the curve frame. An interactive draft-driven deformation (DDD) method is introduced to customize the shoe-last mesh based on contour curve bending. Finally, we propose an encoding/decoding technique to realize global or separate shoe-last reuse in order to share design forms of shoe-last rapidly.

2 Multi-layer parametric definition for customized shoe-last

An MLPD of shoe-last based on the elements of parametric curves and surfaces is proposed to improve the friendly model manipulation, and to meet the requirements of constraints (Table 1 and Fig. 3); six parametric layers are introduced to give descriptions of necessary elements in the shoe-last design. The elements in feature point layer $\{F_{pi}\}$ and dimension layer $\{S_i\}$ can be acquired by the methods in Lu [26]. Here 8 feature points including last forefront point, front land point, etc, are recognized from the shoe-last surface, and then dimension parameters are calculated based on the recognized features.

Feature point layer $\{F_{pi}\}$ and dimension layer $\{S_i\}$ are used to describe feature points and calculate dimensions of shoe-last. Before the recognition of feature points, as the scanned model may not be placed in a proper position in the global coordinate frame, it needs to be adjusted through the sagittal plane which passes the rear of shoe-last and the tip of shoe-last. Then the specific definitions of feature points and dimensions are given in Table 2. The accuracy is related to the adjusted shoe-last position in this coordinate system; several adjustments will improve of accuracy for feature point recognition.

It is defined that the parametric curve S_{pl} consists of control points S_p, discrete lines D_L and curve type T_p (Fig. 4a),

$$\{S_{pl} = (S_p, D_L, T_p) | S_p = \{S_{pi}\}, D_L = \{D_{Li}\}, T_p\},$$
$$D_{Li} = \left(D_{pi}, D_{p(i+1)}\right),$$

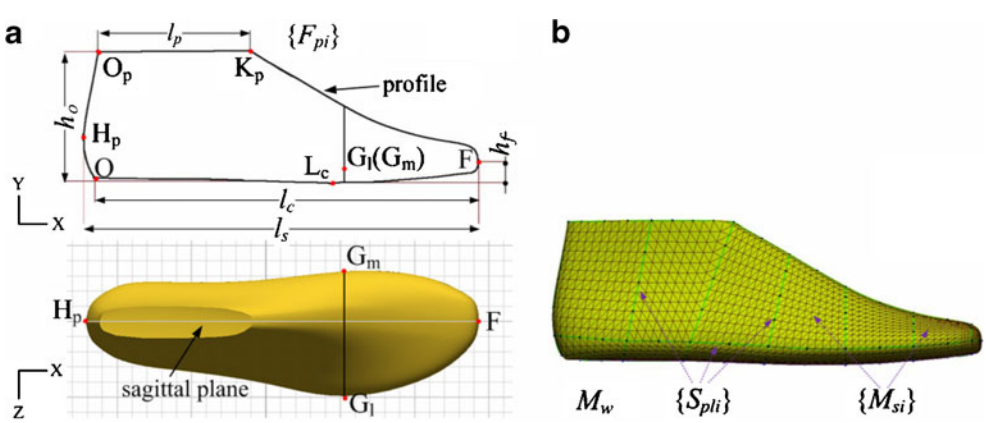


Fig. 3 MLPD of shoe-last

Table 2 Definitions of feature points and dimensions

F	Forefront point of the shoe-last: the point with maximum coordinate value in X direction at the tip of the profile in the toe part of shoe-last.
L_c	Front land point: the point with minimum coordinate value in Y direction on the profile curve.
O	Heel rear point: the point representing the rear of shoe-last with minimum coordinate value of bottom curve of profile in X direction.
H_p	Pternion point: the point representing the posterior point of the heel with minimum coordinate value in X direction on the profile.
O_p/K_p	Posterior point/forefront point of the shoe-last top plane on the profile curve.
G_l/G_m	Lateral/medial point of the ball-toe part giving the width of the shoe-last.
h_o	Heel height: vertical distance from the front land point L_c to posterior point of heel top plane O_p.
h_f	Toe height: vertical distance from the front land point L_c to forefront point of the shoe-last F.
l_p	Heel top length: the distance along x-axis from posterior point O_p to forefront point K_p.
l_c	Last bottom length: the distance along x-axis from heel rear point O to forefront point of the shoe-last F.
l_s	Last full length: the distance along x-axis from Pternion point P to forefront point of the shoe-last F.

where D_{pi} is the discrete point. Parametric curve layer is used to describe contour curves.

It is defined that triangle mesh M consists of vertex $P=\{P_i\}$, edge $E=\{E_i\}$, and triangle $T=\{T_i\}$ (Fig. 4b), where

$$P_i = \{\{AE_i\},\ \{AT_i\}\},$$
$$E_j = \{(P_{j0}, P_{j1}),\ \{AE_j\},\ \{AT_j\}\},$$
$$T_k = \{(P_{k0}, P_{k1}, P_{k2}),\ (E_{k0}, E_{k1}, E_{k2}),\ \{AT_k\}\}.$$

$\{AE_i\}$, $\{AE_j\}$ are respectively the adjacent edge set of P_i and E_j; $\{AT_i\}$, $\{AT_j\}$, and $\{AT_k\}$ are respectively the adjacent triangle set of P_i, E_j, and T_k. Parametric surface layer is used to describe shoe-last mesh.

Finally, the MLPD of shoe-last is presented as $L_p = (\{F_{pi}\},\ \{S_{pli}\},\ \{M_{si}\},\ \{S_i\},\ \{C_i\})$.

3 Interactive draft-driven deformation based on parametric reconstruction

3.1 Piecewise reconstruction

Piecewise reconstruction can simplify and regularize the mesh of shoe-last, which benefits shoe-last surface editing. The whole shoe-last is divided into many regions according to feature points presented in MLPD. By using the slice technique, the shoe-last mesh can be reconstructed with regular vertex arrangement. And as the region division is based on the feature points, the following surface manipulations will be closely related to the shoe-last structure, so it can enhance the rationality for shoe-last design.

According to the MLPD of shoe-last, we take the elements of feature point layer as reference and divide the surface into many regions. It reflects the following requirements:

The shoe-last should be divided into several parts in accordance with recognized feature points so as to manifest the structure of shoe-last, so that it will benefit the following surface editing.

It should support the method of reconstructing shoe-last with slice technique [27].

Specific division should be made where the local curvature changes drastically in order to decrease the difference between the reconstruction model and the original one.

According to the requirements above, we divide a shoe-last into 14 regions with three general parts (Fig. 5): heel, instep–waist, and ball–toe. Some auxiliary points are added to help make this partition. K_c is used to differentiate the heel part and instep–waist part, which is inserted by proportion with respect to last full length. The forefront part and rear part of shoe-last are divided according to F_c and H_c, respectively, for specific treatment. F_c is the given by proportion differed by shoe-last type with respect to full length. The region R_{fl} and R_{fm} are divided by the section across F_c and profile curve. These two regions are used to simulate the tips in different types of shoe-last. H_c is the midpoint of O and K_c on the profile curve, and H_p is the midpoint of O_p and K_p on the

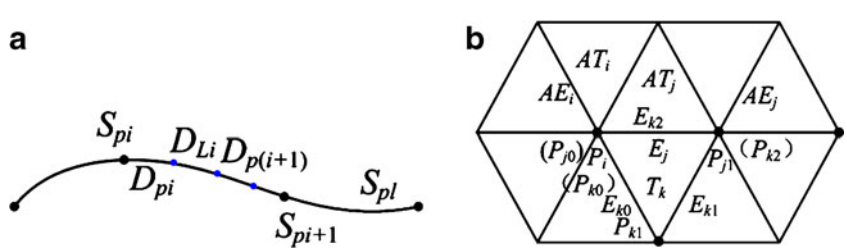


Fig. 4 Element definitions of parametric layers

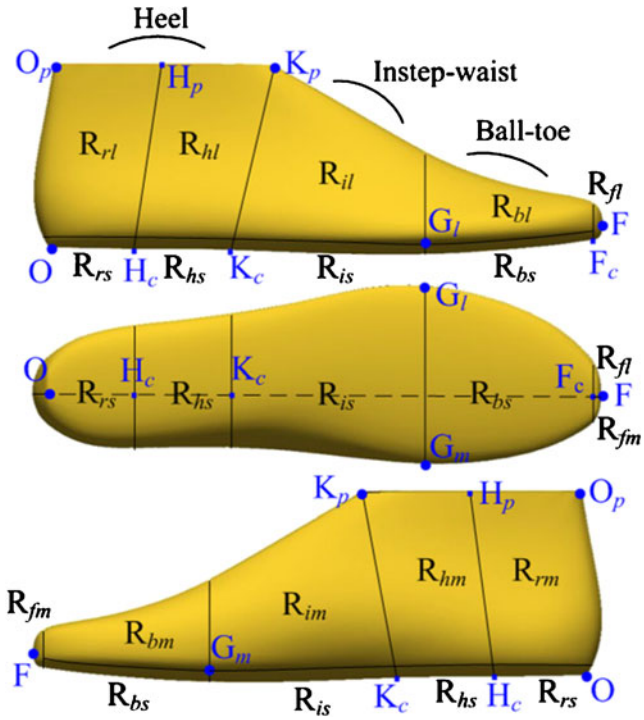


Fig. 5 Surface division of shoe-last

profile curve. Then the sole outline of shoe-last divide heel part into six regions (R_{rl}, R_{hl}, R_{rs}, R_{hs}, R_{hm}, R_{rm}). In order to manifest topology of the piecewise surface, we classify these regions into two categories:

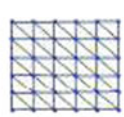
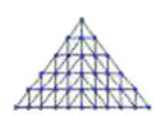
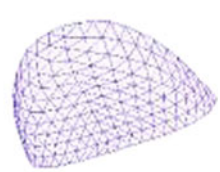
The region similar to quadrilateral, as region R_{rl}, R_{hl}, R_{il}, R_{bl}, R_{hs}, R_{is}, R_{bs}, R_{rm}, R_{hm}, R_{im}, R_{bm};
The region similar to triangle, as region R_{rs}, R_{fl}, R_{fm};

We employ the slice technique [27] to generate piecewise surfaces based on the original shoe-last surface. The intersection nodes acquired by shoe-last slicing are sampled and arranged in an equal-distance array, and they are connected with different topological structures (Table 3) to create regularized piecewise surfaces represented by parametric surface layer $\{M_{si}\}$. Then the adjacent piecewise surfaces are connected with each other to create a whole shoe-last surface. However, the connecting process cannot guarantee the smoothness near the boundaries of piecewise surfaces. It is necessary to introduce smooth treatment to ensure the quality of the shoe-last surface. The existing approaches can acquire good results. We employ Laplacian-smoothing treatment to solve the possible problem, and convert the surface into a smooth one. Figure 6 shows the original scanned shoe-last (Fig. 6a) and its reconstruction surface (Fig. 6b) of which the mesh vertex are in order.

The slice methods, such as the slice direction as well as the slice distance which are decided by the shapes of piecewise surfaces, are differed by regions so as to enhance the precision of the reconstruction. Thus, the details of the original surface can be mainly reserved by adopting the slice technique. The dimensions such as length and girth which are important for shoe-last design and manufacturing may have slight change in the process. So the error rate of shoe-last is defined as the evaluation of the dimension alterations. It is defined as the mean error of dimensions:

$$E(M_o, M_r) = \frac{1}{N}\sum_{i=1}^{N}\frac{|D_o(i) - D_r(i)|}{D_o(i)} \tag{1}$$

Table 3 Topologic structure

Region type	Similar to quadrilateral	Similar to triangle
Region	R_{rl}, R_{hl}, R_{il}, R_{bl}, R_{hs}, R_{is}, R_{bs}, R_{rm}, R_{hm}, R_{im}, R_{bm}	R_{rs}, R_{fl}, R_{fm}
Topologic structure	 Quadrilateral topology	 Triangle topology
Piecewise surface example	Surface-piece R_{rl}	 Surface-piece R_{rs}

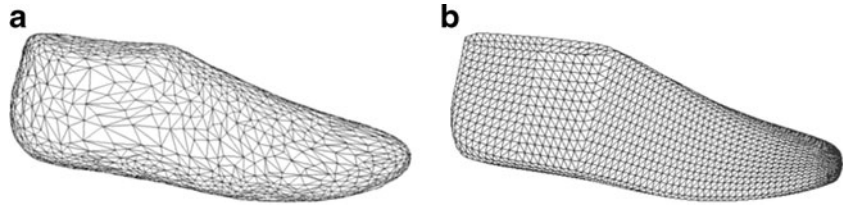


Fig. 6 Shoe-last reconstruction

where, $D_o(i)$ and $D_r(i)$ are dimensions of original mesh and reconstructed mesh, respectively. N is the number of shoe-last dimensions.

3.2 Extraction of contour curves

Contour curves are generated from the boundaries of every piecewise region. They are represented by the parametric curve definition and can be used for the deformation of surfaces. We can uniformly extract the control points from the vertices for they are arranged regularly after piecewise reconstruction. For instance, the control points (Fig. 7) on the boundary of region R_{il} and R_{bl} (Fig. 5) are extracted every three-edge distance of the surface. There are many types of parametric curves for choices to create contour curves from these points, such as Bezier curve or B-spline curve. As control points are sampled from the vertices and we hope the contour curves are generated along the surface of shoe-last. Although the interpolation of Bezier or B-spline curve can pass these points, it needs more calculation in this process. A simple solution is to use Ferguson curve passing the control points and it can also achieve good applied results in the contour curve representing (Fig. 7). These curves are attached on the surface of shoe-last forming a curve frame in accordance with the features. Based on the region division of shoe-last surface, there are totally nine curves generated from the control points. Two curves (C_{S2} and C_{S4}) are added specifically to divide the instep–waist, ball–toe region. Thus, this curve frame attached on the shoe-last surface is generated for the further adjusting of shoe-last. These curves can be classified into two categories:

(1) Envelop contour curve (C_{E1}, C_{E2} in Fig. 7) is the contour curve which envelops shoe-last in a 2D view. The control points of curves are sampled from vertices by appropriate distance, then take Ferguson curve as the envelop curve.
(2) Section contour curve (C_{S1}–C_{S7} in Fig 7) is the contour curve extracted from slice of shoe-last. The control points of curves are sampled from vertices on the slice by equal distance, and then generate a Ferguson curve.

3.3 Parametric constraints of contour curves

The contour curves are created from control points, respectively, and they do not share the topologic information with relative adjacent ones. This kind of curve frame

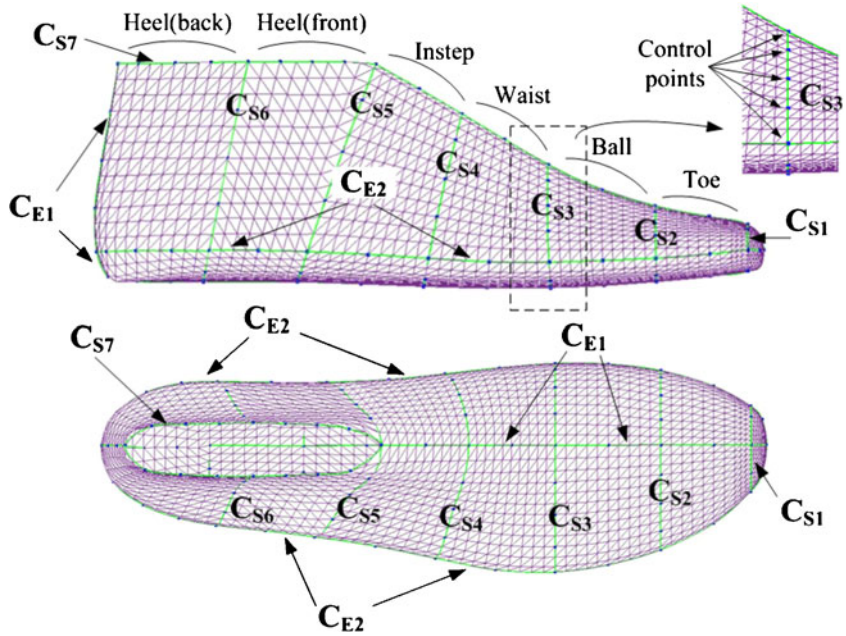


Fig. 7 Parametric contour curves: $\mathbf{C_{E1}}$ the profile curve, $\mathbf{C_{E2}}$ the outline curve, $\mathbf{C_{S7}}$ heel top section curve, $\mathbf{C_{S1}}$–$\mathbf{C_{S6}}$ section contour curves which can divide shoe-last into different parts

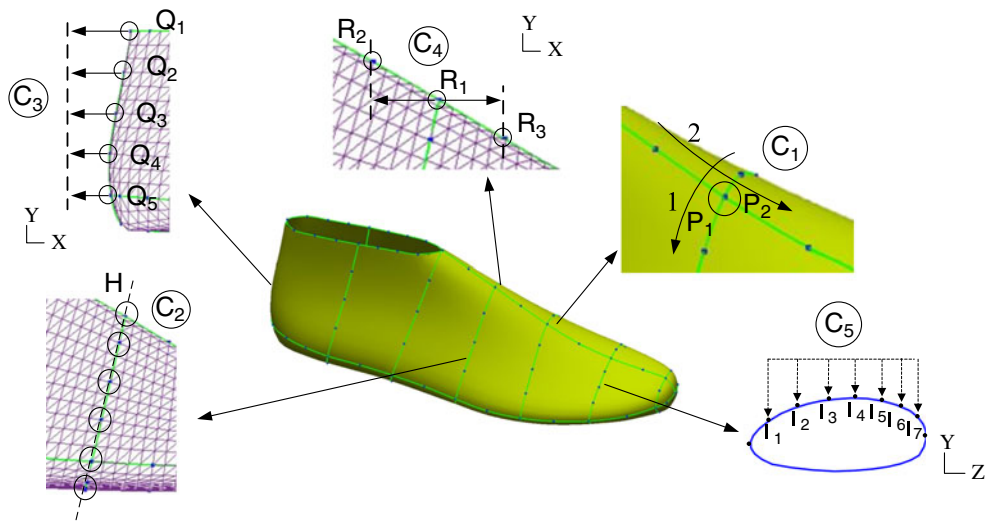


Fig. 8 Constraints in the contour curves

would not keep a stable structure if any of the curves changes. In order to enhance the robustness of this curve frame, parametric constraints are introduced in this paper. They make the curves into a stable system so that it will not crash down when local changes of curves happen. According to the actual situation of curves, here we propose concurrent constraint (C_1), coplanar constraint (C_2), dimension constraint (C_3), etc. (Fig. 8 and Table 4).

We define constraint $C_i = \{(e_1, e_2, \ldots e_i), \text{Hd}\}$, where ($e_1$, e_2,…e_i) is the set of relative control points, Hd is the treating method (Table 4).

Generally, each control point on the curve frame should follow all constraints in Table 4 when its position changed. The constraint C_1, C_2, and C_3 can be defined as "strong constraint" because they should be obeyed strictly, as C_4 and C_5 are comparable "weak" constraints. In order to avoid conflicts between constraints, they are solved in the sequence $C_5 \rightarrow C_4 \rightarrow C_3 \rightarrow C_2 \rightarrow C_1$ so as to strictly meet the strong constraints and sometimes release the weak constraint.

3.4 Draft-driven deformation

3.4.1 General method

Many existing design methods are driven by dimensions to modify the shape of shoe-last [13, 21]. Thought it can

Table 4 Constraint description

Constraint type	Description
Concurrent constraint	$C_1=\{(\boldsymbol{P}_1, \boldsymbol{P}_2), \text{Hd}_1\}$; Hd_1: control points in the set with identical coordinate value. For instance, point $\boldsymbol{P}_1$ is on curve 1 and point $\boldsymbol{P}_2$ is on curve, and they ought to share the same coordinate value because of the intersection of two curves. This constraint will continuously effect whenever one of the two points moves.
Coplanar constraint	$C_2=\{(\boldsymbol{D}_1, \boldsymbol{D}_2, \boldsymbol{D}_3, \boldsymbol{D}_4), \text{Hd}_2\}$; Hd_2: control points in the set with coplanar coordinate value. For instance, the control points on the section plane ought to keep on the same plane (plane H) when any of these points moves.
Dimensional constraint	$C_3=\{(\boldsymbol{Q}_1), \text{Hd}_3; \text{Hd}_3\}$: control points in the set should not exceed a certain precise dimensional limit in x, y, or z direction. For instance, control points $\boldsymbol{Q}_1$–$\boldsymbol{Q}_5$ cannot exceed the limit in x direction, respectively. The limit is defined proportionally by the shoe-last full length. Dimensional constraint is used to avoid excessive and unacceptable modification in the process of deformation.
Directional constraint	$C_4=\{(\boldsymbol{R}_1, \boldsymbol{R}_2, \boldsymbol{R}_3), \text{Hd}_4\}$; Hd_4: control points in the set should not overleap adjacent points along the curve direction. For instance, $\boldsymbol{R}_1$ should not exceed $\boldsymbol{R}_2$ backward or $\boldsymbol{R}_3$ forward in x direction. Directional constraint is used to guarantee the rational shape of contour curves.
Following constraint	$C_5=\{(\boldsymbol{I}_1, \boldsymbol{I}_2, \boldsymbol{I}_3, \boldsymbol{I}_4, \boldsymbol{I}_5, \boldsymbol{I}_6, \boldsymbol{I}_7), \text{Hd}_5\}$; Hd_5: relative elements in the set move in proportion with the select element. For instance, $\boldsymbol{I}_1$, $\boldsymbol{I}_2$, $\boldsymbol{I}_3$, $\boldsymbol{I}_5$, $\boldsymbol{I}_6$, and $\boldsymbol{I}_7$ on section curves move in proportion with $\boldsymbol{I}_4$ when the position of $\boldsymbol{I}_4$ is changed in y direction. For these points (k=1, 2, 3, 5, 6, 7), the displacement of each point in y direction $\Delta Y_k = \Delta Y_4 \cdot \lvert Z_k - Z_4\rvert / \lvert Z_1 - Z_7\rvert$, where ΔY_4 is the displacement of point $\boldsymbol{I}_4$, and s_k is the z-coordinate value of each point.

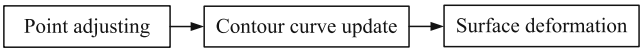


Fig. 9 Procedure of draft-driven interactive deformation

meet the requirements of shoe-last design, it lacks of interaction which could express designer's idea freely. Commercial systems such as Shoemaster and Easylast can provide editing of profile curve and outline curve, but it still needs to operate in 2D view. Some studies [15, 16, 19] proposed local deformation method by interactive manipulation, but they do not systematically combine well with the features of shoe-lasts to guide surface shape modification. Thus we propose an interactive DDD method based on MLPD of shoe-last. It realizes the surface deformation by bending the contour curve frame which is extracted from the piecewise surfaces according to the shoe-last features so as to improve the interaction and intuition in shoe-last design. According to the procedure of DDD (Fig. 9), control point moves lead to contour curves bending, and then the updated curves lead to surface deformation. Therefore, by means of adjusting control points, we can edit relative shoe-last surface.

3.4.2 Deformation process

This paper introduces two approaches to adjust the curves:

(1) Directly drag a control point, and real-time solve constraints on the point to determine its coordinate values, and then update the relative curves (Fig. 10a and b).
(2) Draft a curve by hand, and map relative control points to the drafted curve, then update the contour curves (Fig. 10d and e). This procedure will be elaborated below.

The profile curve (C_{E1} in Fig. 7) and bottom outline curve (C_{E2} in Fig. 7) are the two main curves in the frame,

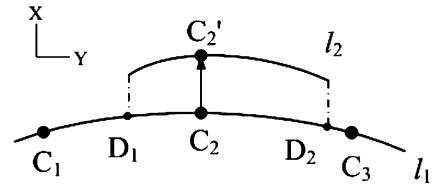


Fig. 11 Mapping of control point

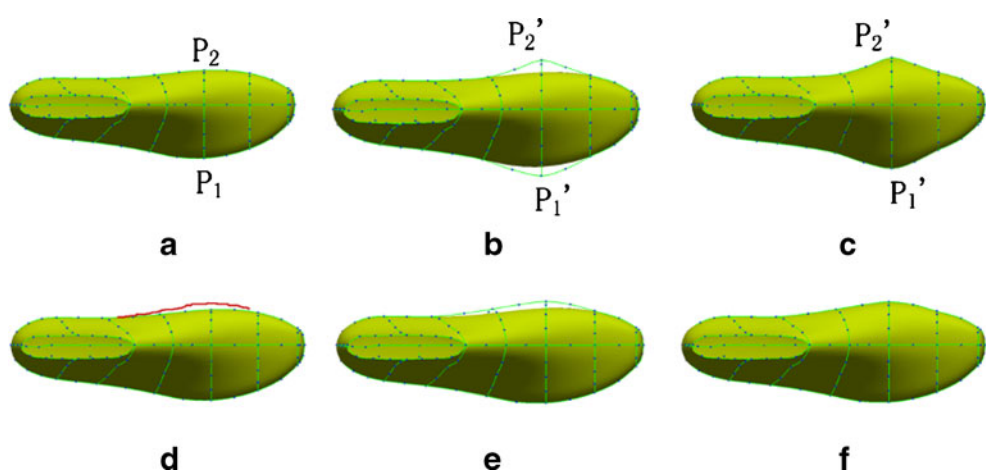
which determines the general shape of shoe-last. By drafting curves in a 2D view and mapping control points to the new curves, we can speed-up the adjusting of contour curve. It can be achieved by following steps:

Step 1 Sketch path pre-processing. Generally, the path drawn by hand with pen input device (such as mouse or digital pen) is not always so smooth because of hand shaking, which could lead to mapping error. We sample points by equal distance on the sketch path and generate a spline curve in the pre-processing so that the path noise can be avoided.

Step 2 Control point mapping. First we select a mapping direction (e.g., normal direction or axis direction), then the domain on the contour curve is decided by mapping sketched curve in a 2D view (Fig. 11, the mapping domain of sketched curve (l_1) is between point D_1 and D_2). Sometimes the profile curve or the bottom contour curve is not a planar curve, so it is necessary to remain the coordinate value on the direction vertical to the 2D view plane in the process of mapping (Fig. 11, control point C_2 is in the mapping domain, so we project it to sketched curve l_2, and update it to C_2' with the same coordinate value on z-axis).

Step 3 Constraint check. All control points are affected by constraints. In order to keep a robust curve frame, we also need to follow the priority of constraint solving and to check the relative

Fig. 10 The process of draft-driven deformation



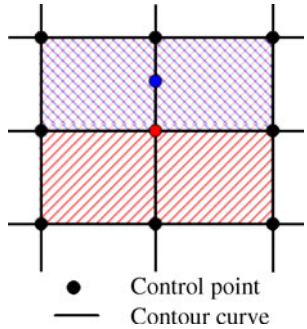


Fig. 12 Deformation ROI

control points. If the updated control points meet the requirements of constraints, then the curves get updated; else we abandon this sketched curve.

The local surfaces enveloped by contour curves in this paper are quadr-boundary surfaces. The deformation region of interest (ROI) is based on its corresponding control point. The moving of control point affects its adjacent quadr-boundary surfaces (e.g., in Fig. 12, the center control point affects four surface regions in red line and the control point in blue affects two adjacent regions in blue). After the adjusting of contour curves, vertices on the boundary of piecewise surfaces are actually updated. In order to preserve the details of the surface we adopt a method similar to u–v coons interpolation but it focuses on the increment led by the control point adjusting (Fig. 13). If original corner points are $\boldsymbol{P}(0,\ 0)$, $\boldsymbol{P}(0,\ 1)$, $\boldsymbol{P}(0,\ 1)$, $\boldsymbol{P}(1,\ 1)$, and original boundaries of curves are $P(u,\ 0)$, $P(u,\ 1)$, $P(0,\ v)$, $P(1,\ v)$, the vectors $\boldsymbol{\Delta}(u,\ 0)$, $\boldsymbol{\Delta}(u,\ 1)$, $\boldsymbol{\Delta}(0,\ v)$,$\boldsymbol{\Delta}(1,\ v)$ can be easily acquired after the adjusting of contour curves, then the displacement $\boldsymbol{\Delta}(u,\ v)$ of $\boldsymbol{P}(u,\ v)$ can be calculated by the following formula.

$$\Delta(u,v) = \lambda 1 \cdot [u \quad 1-u] \cdot \begin{bmatrix} \Delta(0,v) \\ \Delta(1,v) \end{bmatrix} + \lambda 2 \cdot [v \quad 1-v] \cdot \begin{bmatrix} \Delta(u,0) \\ \Delta(u,1) \end{bmatrix}, \quad \text{where } \lambda 1 + \lambda 2 = 1 \tag{2}$$

And if λ1=λ2=0.5, the vectors on the u direction and the v direction have the same influence to $\boldsymbol{\Delta}(u,\ v)$. Thus we can get the new value of $\boldsymbol{P}(u,\ \mathrm{v})$ after the curve changing

$$P'(u,\mathrm{v}) = P(u,\mathrm{v}) + \Delta(u,v) \tag{3}$$

The new value $\boldsymbol{P'}(u,\ \mathrm{v})$ is not the result of the updated boundary interpolation, but the incremental result based on the previous value of $\boldsymbol{P}(u,\ \mathrm{v})$. With constraints, it can rationally transfer the deformation magnitude from the contour curves to the surfaces in its ROI. So either by control point dragging (Fig. 10b and c) or by contour curve sketching (Fig. 10e and f), the local surface shape can be manipulated in accordance with the contour curves bending. In addition to the manipulation in 2D views, the curves also can be adjusted in proper custom view and the relative surface is affected accordingly (Fig. 14). Thus, the user can modify the surface of shoe-last by adjusting not only the envelop curves but also section curves around the surface in any angle of view. Finally, the surface on the boundaries of the ROI is continuous, but in order to keep the smoothness of the surface; finally, the smoothing process is introduced to ensure the smoothness in the region near the contour curves.

3.4.3 Dimension control

Though the hand contour curve dragging improves the efficiency of local surface deformation in an interactive and intuitive way, it lacks the accurate control to key dimensions (e.g., girth). Therefore, it is necessary to employ the size adjusting method [26] to acquire precise shoe-last dimensions. The feature dimensions such as girth and last full length are important for shoe-lasts but they cannot be adjusted to accurate values during the interactive deformation. We need to use fine adjustment after hand adjusting of shoe-last surface to achieve the size accuracy (Fig. 15).

4 Parametric reuse based on encoding/decoding

Shoe-last is a kind of free-form closed surface which has a unique shape, so it is hard to copy only by local surface deformation. In shoe-last design process, we can greatly reduce design time cost and increase ease of design use by fully reusing existing design style of shoe-last. Thus, we introduce a parametric reuse method based on encoding/decoding technique in order to share design style globally and separately.

4.1 Global parametric reuse

In the real shoe-last design case, sometimes we need to keep the main feature dimensions of current shoe-last

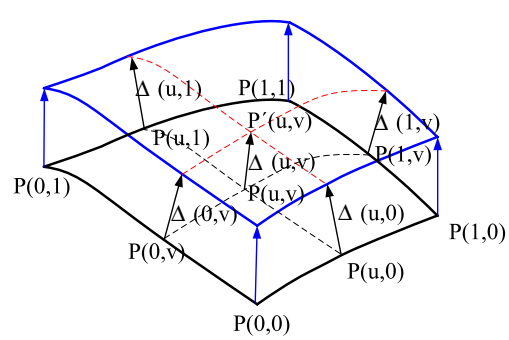


Fig. 13 Surface interpolation

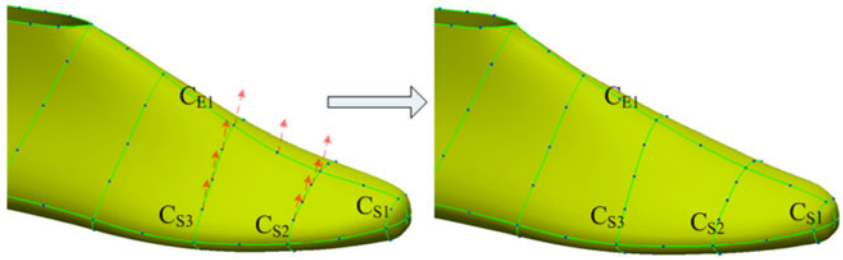


Fig. 14 Local deformation based on DDD: the control points are dragged in *arrow* directions, and relative surface in the front part is modified

model, and copy the existing shoe-last style to the current one. Thus, we do not need to generate a shoe-last initially but achieve shoe-last design based on existing style. In this paper, shoe-last reconstruction models share the same structure of piecewise surfaces, and the topology of each piecewise surface is also identical to the corresponding one. So we parametrically encode shoe-last mesh of existing one to reserve its geometry information, and then reuse it on the current one by decoding, to gain a similar shoe-last established on the original one.

The encoding is based on the feature points described in feature point layer $\{F_{\mathrm{pi}}\}$ in order to reflect the geometry of existing shoe-last. In this paper, we take points (Fig.3a) $\boldsymbol{F}$, $\boldsymbol{O}$, $\boldsymbol{O_p}$, $\boldsymbol{G_l}(\boldsymbol{G_m})$ as reference points to express the encoding method.

We encode the relative position of each vertex of the shoe-last mesh by recoding it as

$$\mathrm{Code}(V_i) = \{u_i, v_i, w_i\},\ \text{where } u_i = \frac{X_i - X_P}{L_{\mathrm{ref}}},\quad v_i = \frac{Y_i - Y_O}{W_{\mathrm{ref}}},\ w_i = \frac{Z_i - Z_O}{H_{\mathrm{ref}}} \tag{4}$$

$$L_{\mathrm{ref}} = |X_F - X_O|,\ W_{\mathrm{ref}} = |Y_{Gl} - Y_{\mathrm{Gm}}|,\quad H_{\mathrm{ref}} = |Z_{Op} - Z_O| \tag{5}$$

L_{ref} is the length of the shoe-last bottom, calculated by the x-axis distance between point $\boldsymbol{A}$ and $\boldsymbol{C}$; W_{ref} is the width of shoe-last toe, namely the y-axis distance between $\boldsymbol{G}$ and $\boldsymbol{G'}$. H_{ref} is the height of shoe-last back, the z-axis distance between $\boldsymbol{C}$ and $\boldsymbol{E}$.

In the process of decoding, we adopt the same reference points of current shoe-last, and decode the code of vertex $\mathrm{Code}(V_i)$ by the following equation:

$$\begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} = \begin{bmatrix} L_{\mathrm{ref}} & 0 & 0 \\ 0 & W_{\mathrm{ref}} & 0 \\ 0 & 0 & H_{\mathrm{ref}} \end{bmatrix} \cdot \begin{bmatrix} u_i \\ v_i \\ w_i \end{bmatrix} + \begin{bmatrix} X_F \\ Y_O \\ Z_O \end{bmatrix} \tag{6}$$

4.2 Separate parametric reuse

Shoe-last can be generally divided into two parts as front part (instep, waist, ball, and toe) and heel part (Fig. 5). Usually, the front part of shoe-last determines the style of shoes and the heel part relates to the fitting of shoes. So most shoe-last deformation is taken on the front part of shoe-last; meanwhile, the back part is always kept in a standard shape. According to the region division, we encode/decode the front part and reuse it separately into a new shoe-last. We adopt surface interpolation method to maintain the smoothness in the transition interval (the instep-waist region in Fig. 5). It can be divided into the following steps:

Step 1 Encode the existing shoe-last toe part with the method in Section 4.1: choose the feature points on the toe part as reference points and encode the vertices to $CodeH(\boldsymbol{V_i})$.

Step 2 Preserve the vertices position of the existing shoe-last in the transition interval.

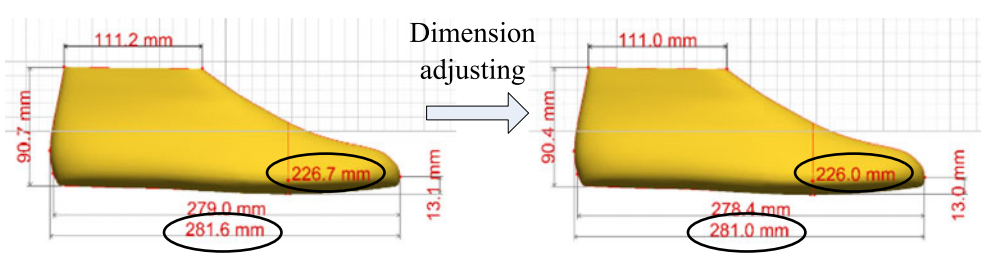


Fig. 15 Dimension adjusting: the full length and toe girth are adjusted by precise dimension values

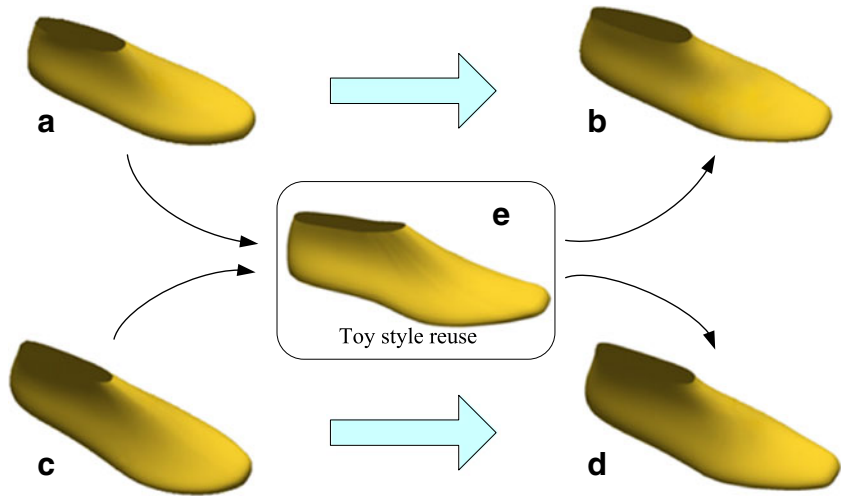


Fig. 16 The front part of last (*e*) is encoded as reuse file in last library. The shoe last (*a*) and (*c*) can separately use the style of shoe-last (*e*) to get new lasts (*b*) and (*d*)

Step 3 Decode $CodeH(\boldsymbol{V_i})$ and release the position information to a new reconstructed shoe-last toe part, then recode the position of vertices in the transition interval as $\{\boldsymbol{V_i'}\}$.

Step 4 The mesh may not match smoothly near the division surface after step 3, so we need to take measures to handle this problem: let N_{seg} be the number of section loop in the transition interval on the mesh, and the vertices in the transition interval are calculated by the following formula:

$$V^{+}{}_{i} = V_{i(n)} \cdot \frac{n}{N_{\text{seg}}} + V'_{i(n)} \cdot \left(1 - \frac{n}{N_{\text{seg}}}\right), \qquad (7)$$

$$n = 0, 1, \ldots, N_{\text{seg}}$$

where $\boldsymbol{V_{i(n)}}$ and $\boldsymbol{V'_{i(n)}}$ are respectively the vertex on the first n-loop of $\{\boldsymbol{V_i}\}$ and $\{\boldsymbol{V_i'}\}$. Figure 16 demonstrates the course and the result of the reuse operation.

5 Design and manufactured results

We can conveniently realize the interactive shoe-last design by adopting the methods presented in the paper.

The scanned shoe-last (Fig. 17a) has an irregular mesh structure. After piecewise reconstruction, we get a regular structure mesh (Fig. 17b) and its shade rendering model (Fig. 17c). The respective error rate is acquired by evaluating the alteration of each dimension (Table 5) and the mean error rate is calculated by Eq. 1. The deviation

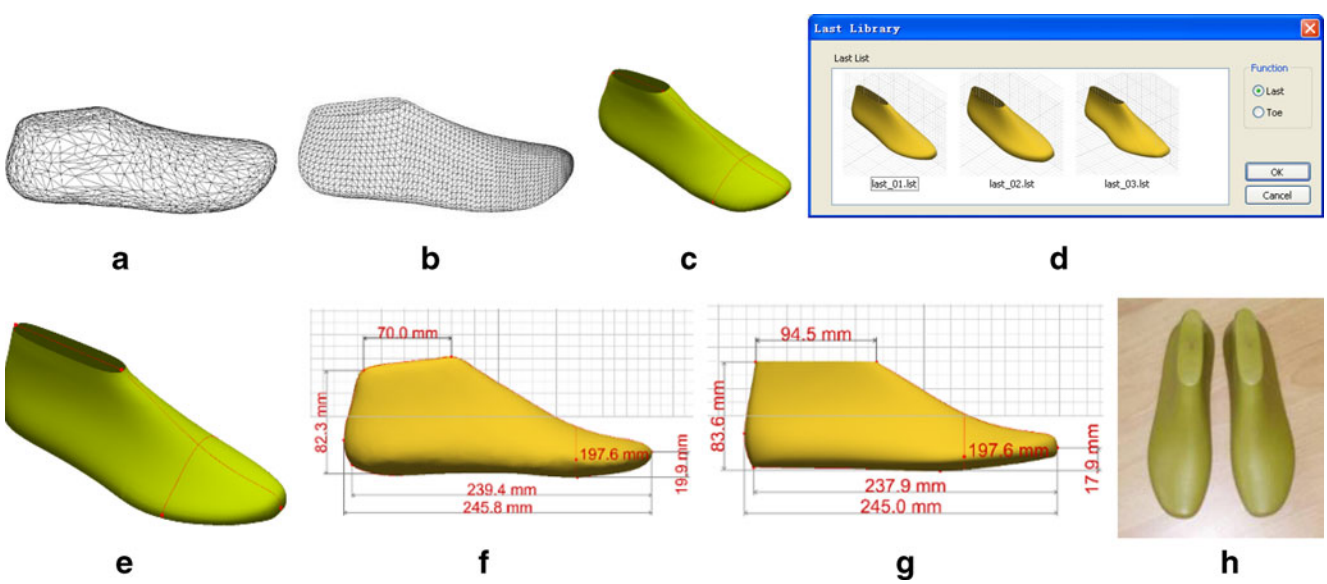


Fig. 17 Piecewise reconstruction and global reuse result

Table 5 Error rate evaluation after piece reconstruction

Dimension	Original mesh (mm)	Reconstructed mesh (mm)	Respective error rate%
Heel height	83.3691	82.0753	1.55
Full length	245.820	245.619	0.08
Sole length	239.689	238.465	0.51
Toe height	19.8951	19.7484	0.74
Toe girth	196.183	196.916	0.37

Error rate (mean): 0.65%

occurred in this process can be revised by using dimension control method (Section 3.4.3) to maintain the key dimension parameters. By choosing the last global reuse file (the left last file in Fig. 17d) which has saved in the last library, we get a new shoe-last (Fig. 17e) based on the key dimensions of original last but with the style of the saved last. Figure 17f shows dimensions of the original last and Fig. 17g shows the dimensions of the new last after the reuse operation. It is easy to find that they both share the similar key dimensions, e.g., girth and length, and the shape of the original last has changed similarly to the shoe-last of the reuse file. Figure 17h shows the manufactured result of this reused last.

Then we use this created new last to demonstrate the course of deformation. The contour curves are extracted from the reused shoe-last, which is shown in Fig. 18a and b. The original curve frame (Fig. 18c) can be adjusted to the one in Fig. 18d by interactive curve bending. It leads to the shaping of a new last (Fig. 18e and f). (Please note that figures from Fig. 18b–e are for demonstration; the curve bending drives the surface deformation in real time.) The dimensions of the new last usually must be different, e.g., the girth is obviously changed if we compare the dimension before the deformation (Fig. 17g) with the one after deformation (Fig. 18g). The dimension control interface (Fig. 18h) provides users to input six dimensions like heel height, toe girth, shoe size, etc. The finally dimensions are shown in Fig. 18i and Table 6. Figure 18j is the manufactured model of the last.

We take a new shoe-last from the last library (Fig. 19a) to demonstrate the course of shoe-last separate reuse. We select the saved toe style file in the interface (the right on in Fig. 19b). It is saved from the last in Fig. 19c. Then the system automatically generates a new last which has the toe part style of shoe-last in Fig. 19c with the back part of shoe-last in Fig. 19a. Figure 19e shows its dimensions and Fig. 19f shows the model manufactured from the design case.

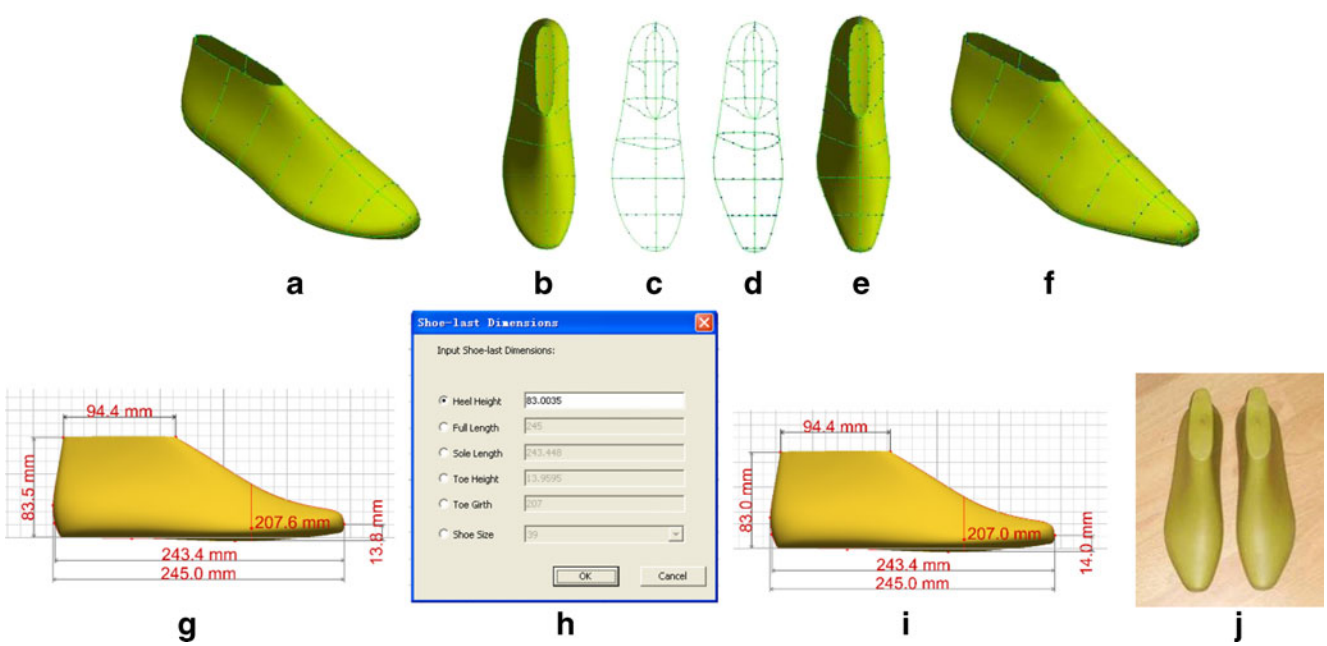


Fig. 18 Interactive deformation and its applied result

Table 6 Adjusting dimensions

Dimension	Before adjusting (mm)	After adjusting (mm)
Heel height	83.5278	83.0035
Full length	245.000	245.000
Sole length	243.448	243.448
Toe height	13.8171	13.9595
Toe girth	207.6	207.000
Shoe size	39	39

6 Discussion and conclusions

The study has provided parametric design methods supporting shoe-last rapid customization. It involves three parts of parametric design concepts: parametric definition of shoe-last for customization, rapid parametric shoe-last customized deformation, and parametric form reuse of shoe-last for rapid design. An MLPD of shoe-last is proposed to support the design method for customized shoe-last. Parameters in certain layers are used to complete the reconstruction of shoe-last; contour curves which are presented by parametric curve layer are extracted from the surface to form a deformation frame. The interactive DDD with parametric constraint solving can speed up the shape design of shoe-last as well as keep the precise control of key dimensions. Finally, parametric form reuse method is proposed to realize the shoe-last reuse to quickly generate a similar new last.

This innovative design system has established a stable parametric shoe-last platform for shoe-last reconstruction and its customized design.

(1) Rational surface division according to shoe-last structure is proposed in this paper to realize the piecewise surface remodeling. The curve frame which consists of contour curves is created to introduce direct and effective manipulation for surface editing.
(2) Five types of constraints are proposed to the control points on the curves to build a stable and robust curve frame. The draft-driven deformation method is realized on the contour curve manipulation with the handling of different constraints.
(3) The global shoe-last form reuse as well as separate shoe-last form reuse method is introduced in this paper to support the rapid shoe-last design with innovation and improve the efficiency in the design process.

The whole adjusting process is visible, intuitive,m and interactive. The design cases and manufacturing results indicate that this new approach is effective.

There are several limitations which leads the future research.

(1) The deformation of the shoe-last is based on the contour curve frame; it may not fulfill the task of adjusting the shape of sub-region local surface. It might be helpful to improve the deformation with more accurate surface manipulation.
(2) The deformation interpolation is a kind of linear interpolation of displacements. There could be a more detail preserving way by improving the interpolation method.
(3) The form reuse method may not match shoe-last well between shoe-lasts of which are in great difference, e.g., a low heel height last and a high heel height last. How to improve the general adaption of the reuse method needs more further study.

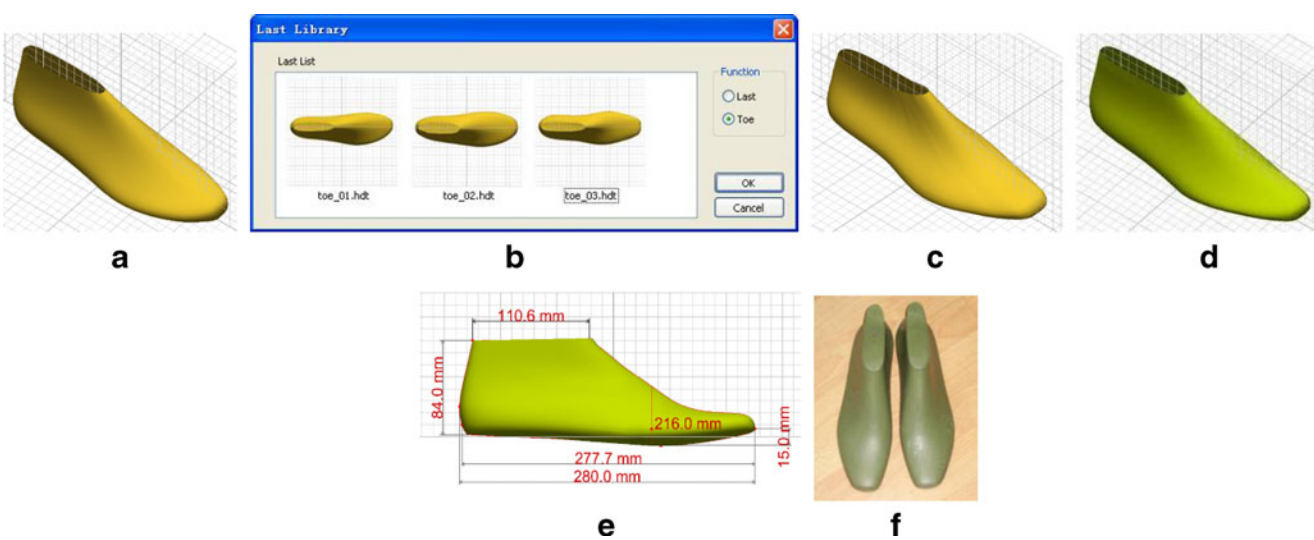


Fig. 19 An example of shoe last separate reuse

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References

1. Barnett L, Rahimifard S, Newman S (2004) Distributed scheduling to support mass customization in the shoe industry. Int J Comput Integ M 17(7):623–632. doi:10.1080/09511920412331292628
2. Du X, Jiao J, Tseng MM (2006) Understanding customer satisfaction in product customization. Int J Adv Manuf Technol 31:396–406. doi:10.1007/S00170-005-0177-8
3. De Rossi N, Rossi A, Trevisani A, Zanotto V (2002) Development and performance assessment of a laser triangulation scanner for digitizing shoe lasts. In Proceedings of AMST 2002, Udine, Italy, July 2002, pp 553–560
4. Lin J, Jiang K, Liu B, Huang C (2008) Rapid registration for 3D data with overlapping range based on human computer interaction. In: Global design to gain a competitive edge. Springer-Verlag London Ltd, Surrey, England, pp 651–659. doi:10.1007/978-1-84800-239-5_64
5. Luximon A, Goonetilleke RS, Tsui KL (2003) Foot landmarking for footwear customization. Ergonomics 46(4):364–383. doi:10.1080/0014013021000045225
6. Zhao J, Xiong S, Bu Y, Goonetilleke RS (2008) Computerized girth determination for custom footwear manufacture. Comput Ind Eng 54(3):359–373. doi:10.1016/j.cie.2007.07.015
7. Cheng F, Perng DB (1999) A systematic approach for developing a foot size information system for shoe last design. Int J Ind Ergonom 25(2):171–185. doi:10.1016/S0169-8141(98)00098-5
8. Mochimaru M, Kouchi M, Dohi M (2000) Analysis of 3-D human foot forms using the free form deformation method and its application in grading shoe lasts. Ergonomics 43(9):1301–1313. doi:10.1080/001401300421752
9. Kos L, Duhovnik J (2002) A system for footwear fitting analysis. Seventh International Design Conference. Croatia, Dubrovnik, pp 1187–1192
10. Witana CP, Feng J, Goonetilleke RS (2004) Dimensional differences for evaluating the quality of footwear fit. Ergonomics 47 (12):1301–1317. doi:10.1080/00140130410001712645
11. Peng W, Xu F, Zhang W (2009) Fuzzy comprehensive evaluation of shoes comfort based on shoe last. International Symposium on Information Proceeding. Huangshan, China, pp 250–253
12. Olivato P, Morricone M, Fubini E, Re A (2007) Foot digitalization for last design and individual awareness of personal foot characteristics. 1st International conference on digital human modeling. Beijing, China, pp 949–958. doi:10.1007/978-3-540-73321-8_107
13. Xiong S, Goonetilleke RS, Witana CP, Au EYL (2008) Modelling foot height and foot shape-related dimensions. Ergonomics 51 (8):1272–1289. doi:10.1080/00140130801996147
14. Leng J, Du R (2005) A deformation method for shoe last customization. Comput Aided Des Appl 2(1–4):11–18
15. Leng J, Du R (2006) A CAD approach for designing customized shoe last. Comput Aided Des Appl 3(1–4):377–384
16. Shi H, Wang B, Jia Q, Luo X (2009) Segment-based adaptive modification for customized shoe last. Optics and Precision Engineering 17(4):880–885
17. Xiong S, Zhao J, Jiang Z, Dong M (2010) A computer-aided design system for foot feature-based shoe last customization. Int J Adv Manuf Technol 46(1–4):11–19. doi:10.1007/s00170-009-2087-7
18. Shi N, Yi S, Xiong S, Jiang Z (2009) A CAD System for Shoe Last Customization. 2009 International Symposium on Applied Computing and Computational Sciences, 24–26 April, Sanya, Hainan Island, China, pp 957–960. doi:10.1109/CSO.2009.229
19. Liu H, Chen L, Ye X, Pan X, Xu Q (2007) Parameterized freeform shape design and deformation. International Conference on Multimedia and Ubiquitous Engineering, Seoul, South Korea, pp 626–632
20. Kim S, Mazumder MMG, Park SJ (2007) A last design with uniform foot pressure free form deformation. 11th international conference on intelligent engineering systems. Budapest, Hungary, pp 89–92. doi:10.1109/INES.2007.4283678
21. Luximon A, Luxinmon Y (2009) Shoe-last design innovation for better shoe fitting. Comput Ind 60(8):621–628. doi:10.1016/j.compind.2009.05.015
22. Yau HT, Chen JS (1997) Reverse engineering of complex geometry using rational B-splines. Int J Adv Manuf Technol 13:548–555. doi:10.1007/BF01176298
23. Kim SH, Shin KH (2009) A method for deforming a surface mmodel using guide surfaces in shoe design. Int J Adv Manuf Technol 42:60–72. doi:10.100./S00170-008-1578-2
24. Hu J, Shen H, Zeng S, Wang Y (2006) B-spline tool offset of a free-form curve in the shoe last high-speed maching CNC system. Int J Adv Manuf Technol 30:864–869. doi:10.1007/S00170-005-0129-3
25. Hwang TJ, Lee K, Oh HY, Jeong JH (2005) Derivation of template shoe-lasts for efficient fabrication of custom-ordered shoe-lasts. Comput-Aided Des 37(12):1241–1250. doi:10.1016/j.cad.2004.12.002
26. Lu G, Chen J, Wang J (2006) Research of shoe last RP based on CAD/CAM integration. Machinery 33(9):28–40, In Chinese
27. Wang M, Lu G, Zhang D (2007) 3D human reconstruction and animation in garment CAD. J Eng Graph 1:1–6, In Chinese